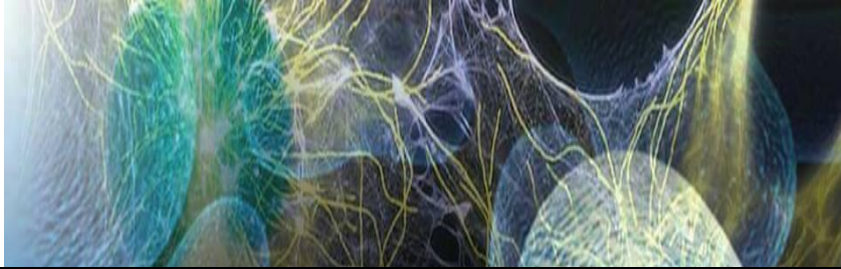


Healing Group

وعزمنّا أن نحسن سعيّا



Isolation of bacteria (Dilution)

2017

Microbiology Lab



Done by: Mohamad BshaiRi

كلية التمريض

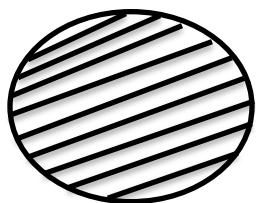


Micro Lab

* Isolation purification and characterization of bacteria *

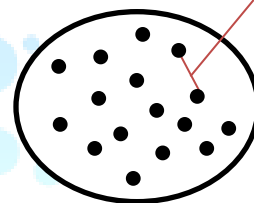
* Isolation of bacteria:

- we use general media (Nutrient agar)



Nutrient agar

Dilation



Different types

Mean isolated colonies

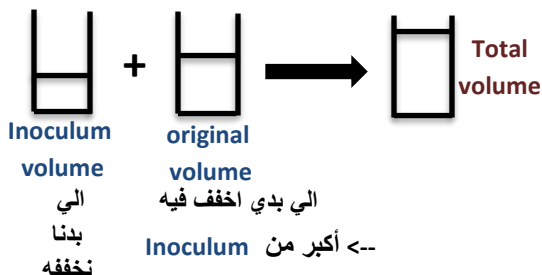
* Two methods for bacteria isolation:

1- serial dilation with pour method

2- serial with spreading methods

Dilution

Normal dilution



*1→ Un dilution

10-fold dilution

شرط يكون من مضاعفات العدد عشرة
(2,5,10,100,1000,....)

*Inoculum

??= 1 ml

0.1 ml

original

9 ml

?? → 9.9

9.99

4.9

1.9

??= 1.2 ml

0.2 ml

3.8 ml

?? → 1.8

4.8

9.8

99.8

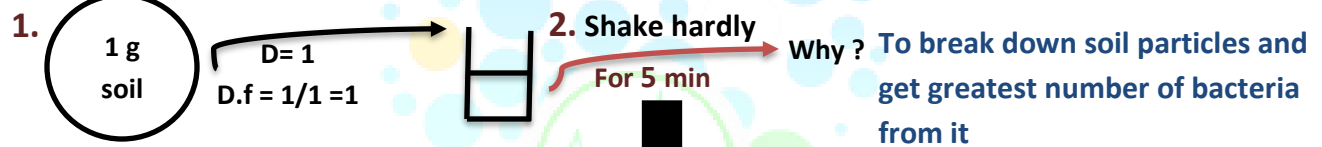
$$\text{* Dilution} = \frac{\text{Previous dilution} * \text{Inoculum volume}}{\text{Total volume}}$$

$$\text{* Dilution factor} = \frac{1}{\text{Dilution}}$$

$$\text{* colony forming until} = \frac{\text{Number of colony} * \text{D.F}}{\text{Inoculum volume}} \quad (\text{C.f.u})$$

* Dilution of source = **1** (means **un diluted**)

* procedure :



Source of bacteria
(most of bacteria
growth in soil is
non-pathogenic)

1 g → 1 ml

$$D = \frac{1 * 1 \text{ ml}}{(99+1) \text{ ml}} = \frac{1}{100} = 10^{-2}$$

(ما اله وحده)

$$D.f = \frac{1}{10^{-2}} = 10^2$$

3. Then put it in bench for **1 min** then take **1 ml** of **st.d.H2o** using sterile pipet put it in test tube (1)
→ (9 ml st.d.H2o)

$$D = \frac{10^{-2} * 1}{10} = 10^{-3}$$

$$D.f = 10^3$$

4. Then mix the tube by **vertex** to give **uniform distribution** of bacteria
5. By new sterile pipet transfer **1 ml** from **1** tube to **2** tubes.

$$D = \frac{10^{-3} * 1}{10} = 10^{-4}$$

$$D.f = 10^4$$

6. Again mix tube **(2)** by **vortex** and add **1 ml** to **3** tubes.

$$D = \frac{10^{-4} * 1}{10} = 10^{-5}$$

$$D.f = 10^5$$

7. Do the previous until you **dilate** it to the **5** tube .

$$D = \frac{10^{-6} * 1}{10} = 10^{-7}$$

$$D.f = 10^7$$

* Surface spreading method :

That is Kept in absolute ethanol

We need **3** agar Nutrient plates , vortex , rotatable plate , **L-shape rod colony counter** (we chose to choose the last **3** test tube)

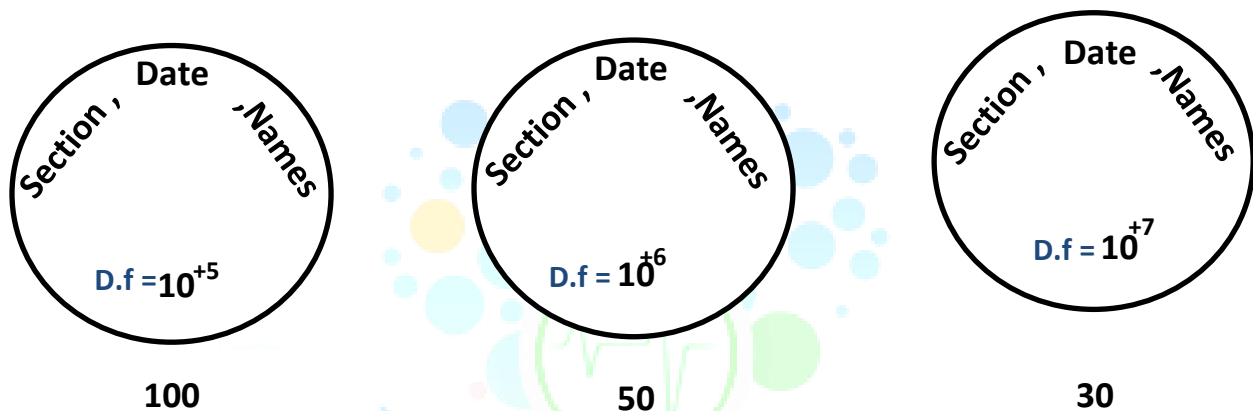
- **Vortex** each test tube and then take and then take **0.1ml** by using new pipet and put in agar plates .

* Then by using **rotatable plate**, put **agar plate** on it , and use **L-shape** to spread the **0.1 ml**

We sterilization it by pass it through heat and cooling
in empty area on agar, then spread (يفضل بخط مستقيم)

* **Incubate it overnight at 37° temp (24 hours)**

* **As dilution factor increase the number of colonies decrease**



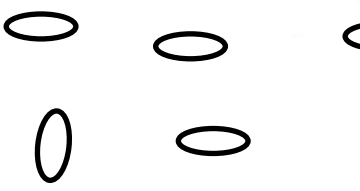
$$Cfu_1 = \frac{100 * 10^{+5}}{0.1 \text{ ml}} = 10^8 \text{ Colony/ ml}$$

$$Cfu_2 = \frac{50 * 10^{+6}}{0.1 \text{ ml}} = 5 * 10^8 \text{ Colony/ ml}$$

$$Cfu_3 = \frac{30 * 10^{+7}}{0.1 \text{ ml}} = 3 * 10^9 \text{ Colony/ ml}$$

$$\text{Cfu average} = \frac{\text{Cfu}_1 + \text{Cfu}_2 + \text{Cfu}_3}{3}$$

Cfu → give estimation for number of bacteria in soil

Size of colony	Number of colony
 Penpoint	↓ 3 - 1000 ↑ contamination uncountable إذا اعلى من ١٠٠٠
 Moderate	↓ 3 - 300 ↑
 Large	↓ 3 - 20 ↑

* **Uncountable** → ما بنحسبها لما نجمع عدد كولوني بالنهاية

* **Contamination** → ما بنحسبها لما نجمع عدد كولوني بالنهاية
كان العدد اقل من ٣ لأنها بتكون ملوثة

* **After count colony, we place plates in fridge**

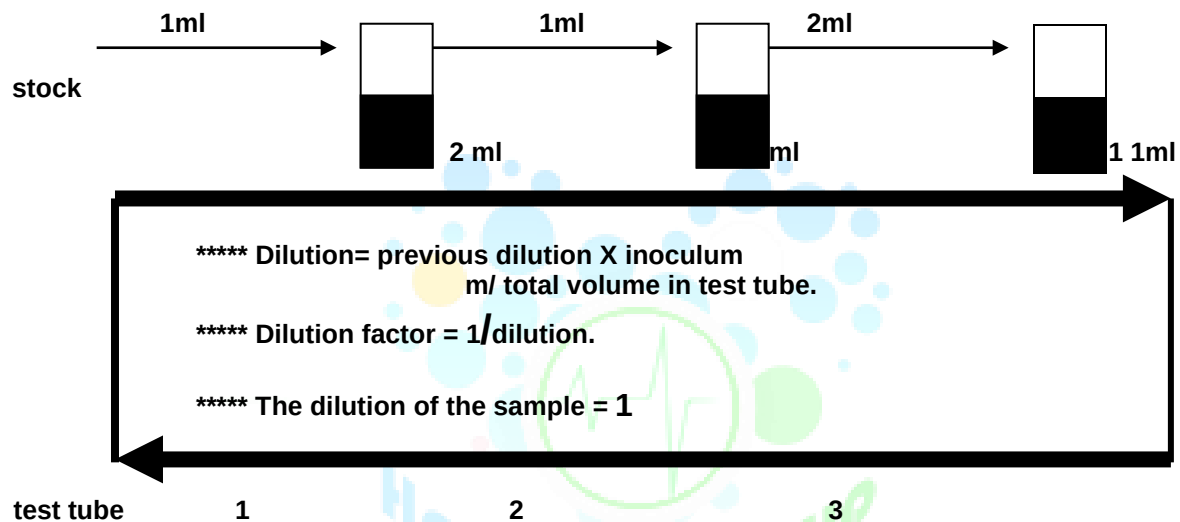
Sheet 1

There are two methods for isolation and purification of bacteria:

- 1- Serial dilution with pour plate method
then Surface streaking method.
- 2- Serial dilution with surface spreading method
then Surface streaking method.

There are two methods for dilution:

1. Normal dilution : e.g.



CALCULATION:

Dilution= previous X inoculum/ total

For test tube 1: $D = 1 \times 1/3 = 1/3$

For test tube 2: $D = 1/3 \times 1/4 = 1/12$

For test tube 3: $D = 1/12 \times 2/3 = 2/36$

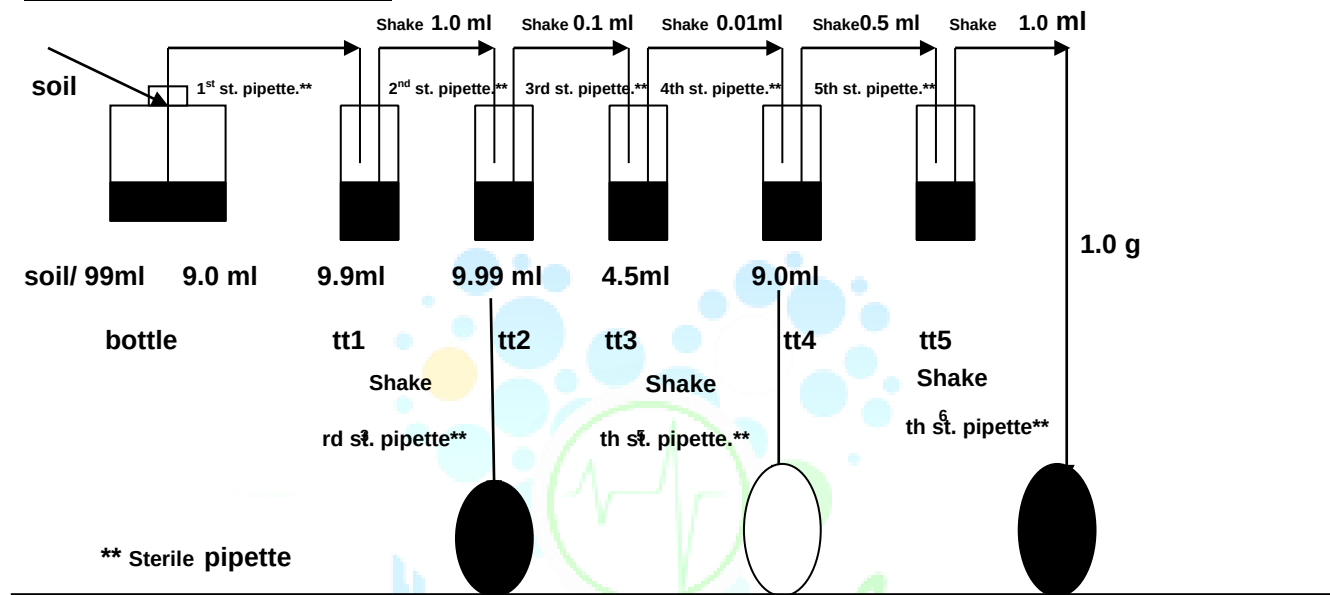
Dilution factor=1/dilution

$DF = 1 / 1/3 = 3$

$DF = 1 / 1/12 = 12$

$DF = 1 / 2/36 = 36/2 = 18$

2- 10- fold dilution



a: Calculation:

Dilution= previous X inoculum / total

In bottle: $D = 1 \times 1 / 100 = 1 / 10^2 = 10^{-2}$

tt1 : $D = 10^{-2} \times 1/10 = 10^{-2} \times 10^{-1} = 10^{-3}$

tt2 : $D = 10^{-3} \times 0.1/10 = 10^{-3} \times 1/100 = 10^{-3} \times 10^{-2} = 10^{-5}$

tt3 : $D = 10^{-5} \times 0.01/10 = 10^{-5} \times 1/1000 = 10^{-5} \times 10^{-3} = 10^{-8}$

tt4 : $D = 10^{-8} \times 0.5/5 = 10^{-8} \times 1/10 = 10^{-9}$

tt5 : $D = 10^{-9} \times 1/10 = 10^{-9} \times 10^{-1} = 10^{-10}$

Dilution factor=1 / dilution

DF = $1/10^{-2} = 10^2$

DF = $1/10^{-3} = 10^3$

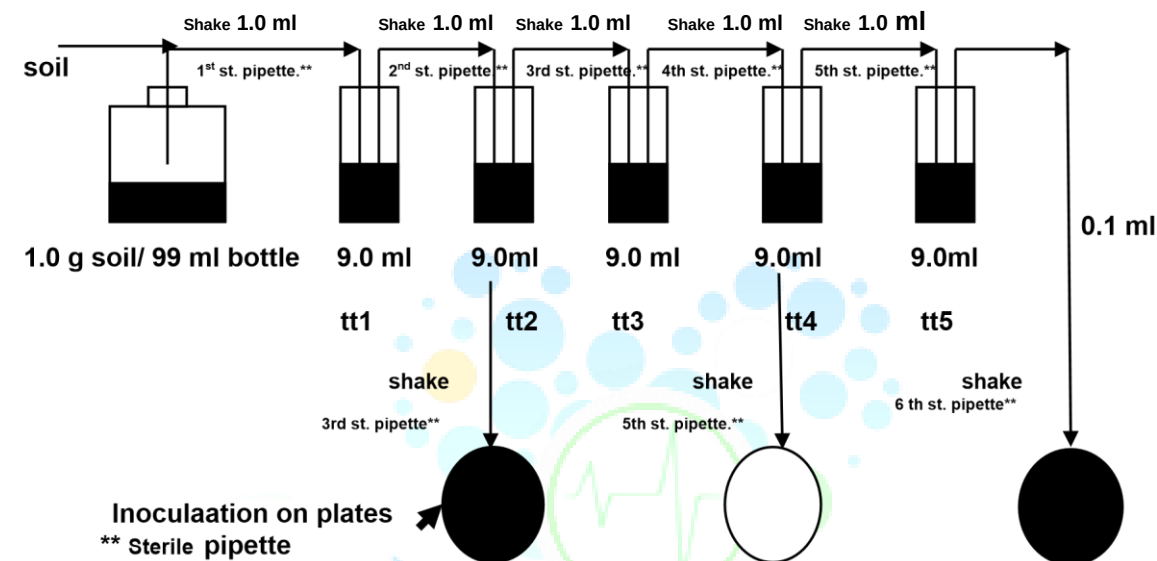
DF = $1/10^{-5} = 10^5$

DF = $1/10^{-8} = 10^8$

DF = $1/10^{-9} = 10^9$

DF = $1/10^{-10} = 10^{10}$

2- 10- fold dilution



a: Caculation:

Dilution= previous X inoculum / total

Dilution factor=1 / dilution

In bottle : $D = 1 \times 0.5 / 50 = 5/500 = 1/100 = 1 / 10^2 = 10^{-2}$

DF = $1/10^{-2} = 10^2$

tt1 : $D = 10^{-2} \times 1/10 = 10^{-2} \times 10^{-1} = 10^{-3}$

DF = $1/10^{-3} = 10^3$

tt2 : $D = 10^{-3} \times 1/10 = 10^{-3} \times 10^{-1} = 10^{-4}$

DF = $1/10^{-5} = 10^4$

tt3 : $D = 10^{-4} \times 1/10 = 10^{-4} \times 10^{-1} = 10^{-5}$

DF = $1/10^{-5} = 10^5$ tt4 :

tt4 : $D = 10^{-5} \times 1/10 = 10^{-5} \times 10^{-1} = 10^{-6}$

DF = $1/10^{-6} = 10^6$

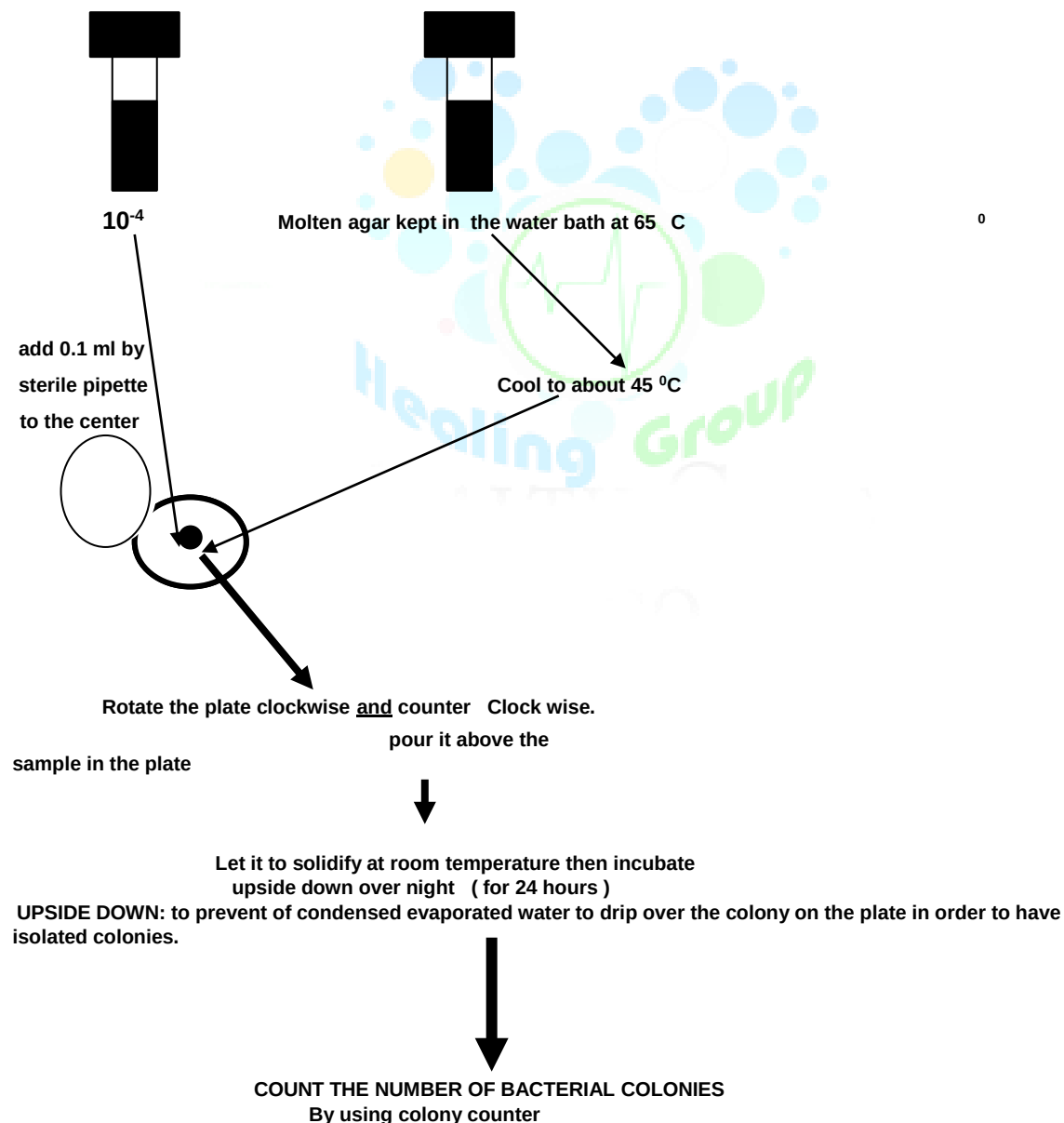
tt5 : $D = 10^{-6} \times 1/10 = 10^{-6} \times 10^{-1} = 10^{-7}$

DF = $1/10^{-7} = 10^7$

Inoculation on plates by two methods:

1. SURFACE SPREADING METHOD: WILL BE USED IN THE LAB.
2. POUR PLATE METHOD:

All plates must be labeled near the edge of the plate:



COUNTABLE PLATE:

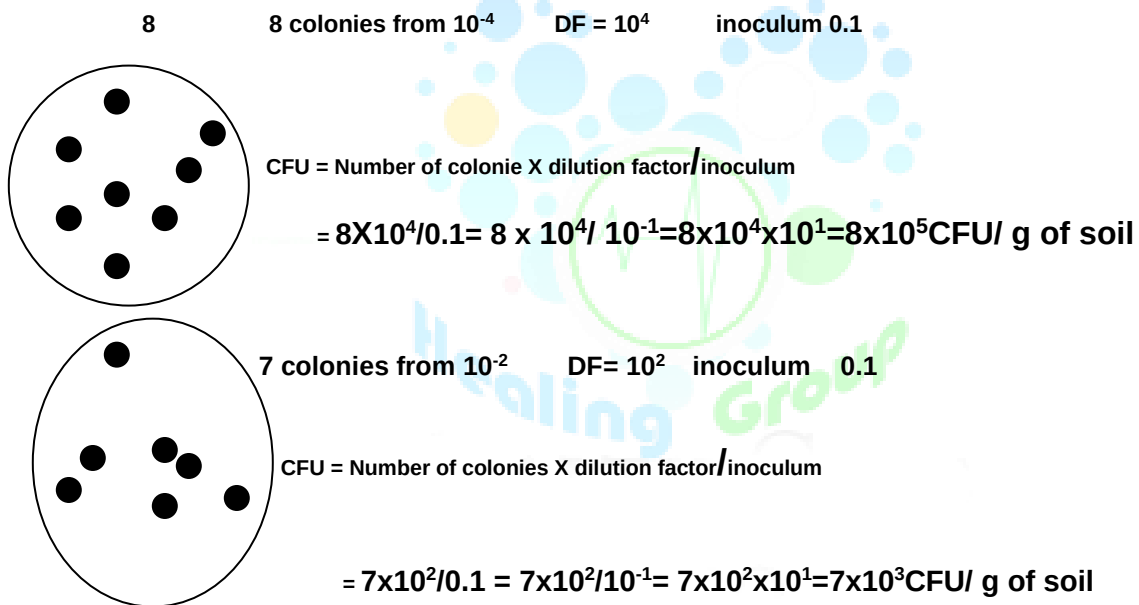
1 – 20 colonies countable for the LARGE colonies

3 – 300 colonies countable for the MODERATE colonies

3 – 1000 colonies countable for the PIN POINT colonies

CALCULATION / GRAM OF SOIL

Counted plates:



Average = $1.6 \times 10^6 + 1.4 \times 10^4 / 2 = 16 \times 10^5 + 0.14 \times 10^5 / 2 = 16.14 \times 10^5 / 2 = 8.7 \times 10^5 \text{ CFU / 1 g of soil}$

**** please forgive us for any mistake!... Healing group



تم بحمد الله